Probabilistic Models of the Visual Cortex

Fall 2026 Homework 3

Prof. Alan Yuille

Instructions

1. **Deadline**: Monday, Nov 23, 2026 at 11:59 PM (Eastern Time).
2. **Late homework**: Late homework will not be accepted unless permission is obtained well in advance for documented extenuating circumstances. Email Prof. Yuille for approval and copy the TAs.
3. **Submission**: one PDF file on Gradescope with your written answers and the plots and answers for Questions 9–11, named `<your_jhed>-hw3.pdf`. Code is not required and will not be graded; a notebook export that shows code is fine (File → Save and Export Notebook As → HTML, then print it to PDF from your browser).
4. **Coverage**: Lectures 15–20 (Question 1 Lecture 15; Questions 2–3 Lecture 16; Question 4 Lecture 17; Question 5 Lecture 18; Question 6 Lecture 19; Question 7 Lecture 19 Ext; Question 8 Lecture 20). 110 points: written Questions 1–8 (72), experimental Questions 9–11 (38).
5. **Warning**: Start the experimental section (Questions 9–11, three notebooks) early! Do not use AI tools (such as ChatGPT, Claude, Gemini or Copilot) to complete this homework. If we detect that AI was used to complete it, you will receive zero points for this homework.
6. **Calculations**: You may use a calculator or Python. Write down the formula and the numbers you plug in, not only the final answer. Give four significant digits; log is the natural logarithm.
7. **Collaboration**: Feel free to discuss the questions with friends, but be sure to write your answers entirely independently!
8. **Questions?** Attend office hours (see the course website) or send an email!

Question 1: Bayes decision theory, deep networks and generative models (8 points)

1. **(3 points) Concepts.**

   (a) Write the formula for the empirical risk $\hat{R}(\alpha)$ of a decision rule α on n annotated examples (I_i, W_i) with loss function L. Why is a network not trained to minimise the 0–1 loss directly, and what loss is minimised instead? **(2 points)**

   (b) The lecture pairs two Gabor filters with phases 0 and $\pi/2$ and takes the square root of the sum of their squared responses, the local energy. Is this a linear operation on the image, like a convolution? Why or why not? **(1 point)**

2. **(2 points) Figure 1: two ways to interpret an image.** In the lower panel of Figure 1 two arrows join the image I and the world state W.

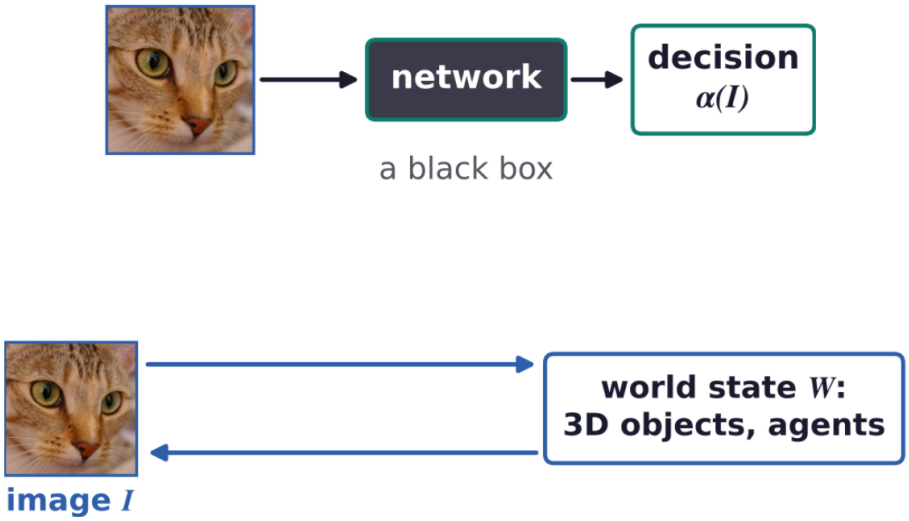


Figure 1: A network (above) and a Bayesian model (below).

(i) Name the two arrows, and the distribution that the arrow from W to I uses. (ii) The lecture calls the network of the upper panel "a black box". What does the arrow back to the image in the lower panel make possible that the black box cannot do?

3. **(2 points) The ventral stream and a failure.**

   (a) The neocognitron alternates two kinds of cells. What do the S-cells and the C-cells do? **(1 point)**

   (b) AlexNet labels a photograph of a cat correctly, but calls a blurred version of it a puffer fish, and calls it a Chihuahua after a crafted perturbation that keeps every pixel within 2 grey levels of the original. Give the one sentence of Bayes decision theory by which the lecture explains both failures. **(1 point)**

4. **(1 point) Diffusing an image into noise.** After t steps of adding a little Gaussian noise, $I_t = \sqrt{\gamma_t}\, I_0 + \sqrt{1-\gamma_t}\, \epsilon$ with $\epsilon \sim \mathcal{N}(0, \mathrm{Id})$ and $\gamma_t = \prod_{s=1}^{t}(1-\beta_s)$. For a constant noise variance $\beta_s = 0.004$, compute γ_{100} and the first step t from which the noise outweighs the image ($\gamma_t < 1/2$).

Question 2: The pinhole camera, vanishing points and the ground plane (8 points)

1. **(3 points) Concepts.**

   (a) What are the two major properties of perspective projection stated in the lecture? **(1 point)**

   (b) The lecture's camera has its pinhole at $\vec{o}$, right-handed unit vectors $\vec{a}, \vec{b}, \vec{c}$ as axes, $\vec{c}$ the direction of gaze, and its imaging plane at distance f behind the pinhole. Write the image coordinates (u, v) of a scene point $\vec{r}$, and name the operation in them that makes the projection non-linear. **(1 point)**

   (c) Scaled orthographic projection approximates perspective projection linearly. What does it lose, and when is it a good approximation? **(1 point)**

2. **(2 points) Figure 2: the ground plane.** Figure 2 shows a camera with a level gaze above a ground plane and three people of equal height at different distances, and below it the image, drawn upright, with two trees.

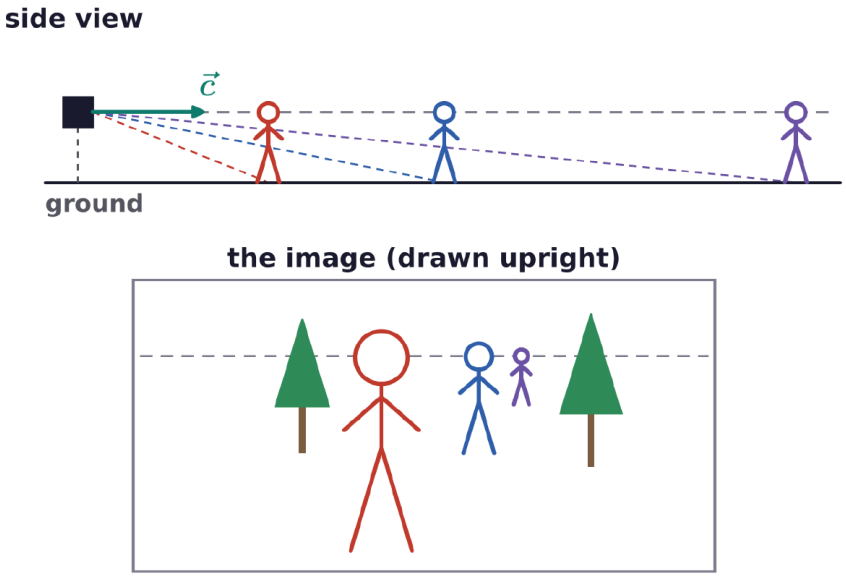


Figure 2: The side view (above) and the image (below).

(i) What is the dashed horizontal line in the image, and why do the three heads lie close to it? (ii) The two trees are vertical lines in the scene. Why do they stay parallel in the image?

3. **(2 points) Depth and height from the ground plane.** A camera 1.4 m above a ground plane has a level gaze and focal length $f = 1000$ pixels; take the image plane in front of the pinhole, so that image distances are positive. (i) A person 1.9 m tall stands at depth $Z = 4$ m: how many pixels below the horizon are the feet, and how many pixels above it is the head? (ii) A second person's feet appear 70 pixels below the horizon and the person's image is 56 pixels tall: what are this person's depth and height, and is the head above or below the horizon?

4. **(1 point) A vanishing point by hand.** The camera of item 1(b), with its imaging plane behind the pinhole, has its pinhole at the origin, $\vec{a} = (1, 0, 0)$, $\vec{b} = (0, 1, 0)$, $\vec{c} = (0, 0, 1)$ and $f = 1$. A line passes through $\vec{r}_0 = (1, 0, 1)$ with unit direction $\hat{\vec{r}} = (1, 1, 1)/\sqrt{3}$. Compute the image (u, v) of the point at distance $\lambda = \sqrt{3}$ from $\vec{r}_0$ along the line, and the vanishing point of all lines with this direction.

Question 3: Stereo, the essential matrix, structure from motion and the Ames room (10 points)

1. **(3 points) Concepts.**

   (a) Two cameras with centres C and C' view a scene point X. Define the epipolar plane, the epipolar lines and the epipoles. **(1 point)**

   (b) With the second camera's coordinates $\vec{X}' = R(\vec{X} - \vec{t})$, $f = 1$ and the image planes in front ($\vec{u} = \vec{X}/X_3$, $\vec{u}' = \vec{X}'/X'_3$), write the essential matrix E in terms of R and $\vec{t}$, and the constraint that corresponding points $\vec{u}$, $\vec{u}'$ satisfy. **(1 point)**

   (c) Why does the $M \times N$ matrix U of the u coordinates of N points in M scaled orthographic views, measured from each view's centroid, have rank at most three? **(1 point)**

2. **(2 points) Figure 3: the Ames room.** In the top view of Figure 3 the orange rays run from the peephole to the far corners of the real room (solid outline); the red and the blue person are of equal height. The right panel is the view through the peephole.

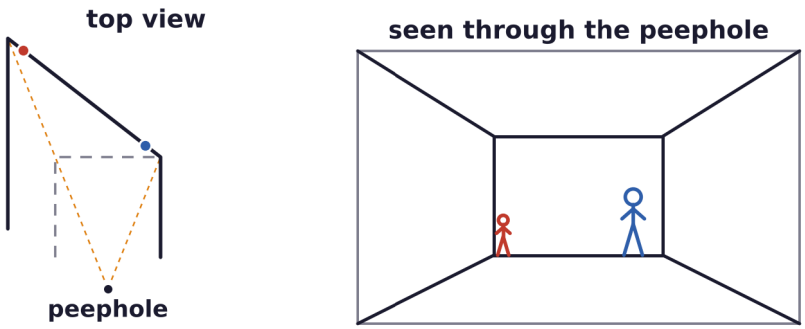


Figure 3: The Ames room.

(i) The room was built by moving the far corners of a room along their rays from the peephole. Why does this leave the view through the peephole unchanged? (ii) Which assumption about the room does the viewer make, and why does it make the red person look smaller than the blue one?

3. **(2 points) Two parallel cameras by hand.** Two cameras have $f = 1$ and image planes in front; the second is displaced by $\vec{t} = (0.3, 0, 0)$ without rotation ($R = I$). Use $([\vec{t}]_\times)_{ij} = \sum_k \epsilon_{ijk} t_k$, so that $[\vec{t}]_\times \vec{X} = \vec{X} \times \vec{t}$. (i) Write $[\vec{t}]_\times$ and E, and the epipolar line $\vec{l} = E\vec{u}$ of the image $\vec{u}$ of $\vec{X} = (0.9, 0.6, 1.2)$: which points of the second image lie on it? (ii) Compute the image $\vec{u}'$ of $\vec{X}$ in the second camera, verify that $\vec{u}$ and $\vec{u}'$ satisfy the constraint of item 1(b), and give the disparity $u - u'$.

4. **(2 points) Structure from motion.**

   (a) The SVD of the stacked $2M \times N$ matrix determines the rows $\vec{K}^m, \vec{H}^m$ of the M affine cameras and the $3 \times N$ matrix of points only up to an invertible 3×3 matrix A. Write this ambiguity, and name the constraints of scaled orthographic projection that reduce it. **(1 point)**

   (b) Why can the size of the scene not be recovered from corresponding points in two views? **(1 point)**

5. **(1 point) Non-rigid motion.** A non-rigid shape is described as a sum of d basis shapes with coefficients that change from frame to frame. What is then the bound on the rank of U, and why?

Question 4: Lambertian reflectance: shadows, photometric stereo and the bas-relief ambiguity (12 points)

1. **(3 points) Concepts.**

   (a) Write Lambert's law for the intensity I of a surface point with albedo ρ lit by a distant light of strength E at angle θ to the surface normal, including the case $\theta > 90°$. **(1 point)**

   (b) The lecture calls Lambertian models "comparatively easy to invert". Which two properties of the model does it give as the reasons? **(1 point)**

   (c) Why do the images of a Lambertian object under several distant lights, without shadows, form a matrix of rank three, and what does a non-zero leftover $\sum_{i \geq 4} D_{ii}^2$ after the SVD of that matrix therefore indicate? **(1 point)**

2. **(2 points) Figure 4: shadows.** Figure 4 shows the lecture's synthetic face lit from the left, with its shadows in blue and orange, and below it a mountain top and a shaft bottom under a sky of lights, with the lights each point receives.

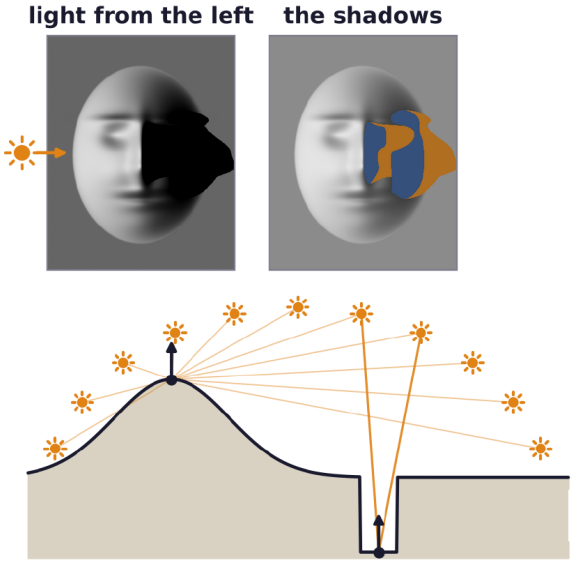


Figure 4: Shadows on the synthetic face (above); a mountain top and a shaft bottom (below).

(i) Which colour marks the attached shadows and which the cast shadows? (ii) The mountain top and the shaft bottom have the same normal and albedo, so the lecture's clipped model, in which each light's contribution is set to zero where the point faces away from it, gives them the same intensity. Which kind of shadow does this model describe, and why is the shaft bottom darker in reality?

3. **(2 points) Photometric stereo by hand.** A pixel with scaled normal $\vec{b} = a\,\hat{\vec{n}}$ (albedo a, unit normal $\hat{\vec{n}}$) is seen, with no shadow, under the distant lights $\vec{s}_1 = (0, 0, 1)$, $\vec{s}_2 = (0.6, 0, 0.8)$ and $\vec{s}_3 = (0, 0.6, 0.8)$ (unit vectors), with intensities $I_j = \vec{b} \cdot \vec{s}_j$ equal to 0.400, 0.464 and 0.212. (i) Solve for $\vec{b}$ and give a and $\hat{\vec{n}}$. (ii) A fourth light $\vec{s}_4 = (-0.96, 0, 0.28)$ is added. What intensity does the clipped model predict for this pixel under $\vec{s}_4$, and why must that image be left out when $\vec{b}$ is estimated by least squares?

4. **(1 point) Convex or concave.** At a pixel the surface $z(x, y)$ has slopes $z_x = 0.3$ and $z_y = -0.4$, so its unit normal is $\hat{\vec{n}}(z) = (-z_x, -z_y, 1)/\sqrt{z_x^2 + z_y^2 + 1}$; the albedo is 1 and the light is $\vec{s} = (0, 0.6, 0.8)$. Compute the intensity $\hat{\vec{n}}(z) \cdot \vec{s}$; then the intensity of the inverted surface $-z$ at the same pixel under the light $\vec{s}' = (0, -0.6, 0.8)$, and under the original light $\vec{s}$. What do the three numbers show?

5. **(2 points) The ambiguity of unknown lights.** With unknown lights, the images fix the scaled normals $\vec{b}$ and the lights $\vec{s}$ only up to an invertible 3×3 matrix A: $\vec{b} \mapsto A^T\vec{b}$, $\vec{s} \mapsto A^{-1}\vec{s}$. (i) Could a better algorithm than the SVD remove this ambiguity from the images alone? Why or why not? (ii) When the normals are required to be those of a surface (integrability), which transformations of the surface $z(x, y)$ survive — write them — and what do they do to the surface?

6. **(1 point) Nine harmonic images.** Why do nine harmonic images suffice to approximate the images of a convex Lambertian object under any lighting?

7. **(1 point) Rendering.** Phong's model lights a surface point by three terms, ambient, diffuse and specular. Which term is Lambert's law, and which depends on where the viewer is?

Question 5: Fitting SMPL to images (10 points)

1. **(2 points) Concepts.**

   (a) What is a body model, and what do SMPL's shape parameters $\vec{\beta}$ and pose parameters $\vec{\theta}$ describe? **(1 point)**

   (b) Name the three steps of fitting a body model to an image by optimisation. **(1 point)**

2. **(2 points) Figure 5: two distance-transform terms.** With S_I the image's silhouette, S_M the model's, and D_I, D_M their distance transforms, the two terms are 1) $\sum_{q \in S_M} D_I(q)$ and 2) $\sum_{q \in S_I} D_M(q)$.

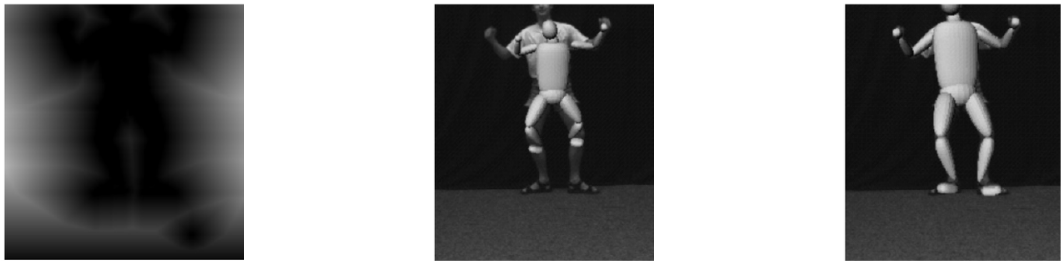


Figure 5: Left, the distance transform D_I of a person's silhouette (dark: near the silhouette); middle and right, two body models fitted to it.

(i) One fit minimised term 1 alone and the other both terms: which is which, and why does term 1 alone allow that fit? (ii) What does term 2 force the model to do, and when is the sum of the two terms zero?

3. **(3 points) One Gauss-Newton step on a planar arm.**

   (a) Fitting minimises $e(\mathbf{x}_t) = \mathbf{e}^T\mathbf{e}$, where $\mathbf{e} \in \mathbb{R}^{2N}$ stacks the differences between the N predicted and the N observed image features. What is the Jacobian $\mathbf{J}_t$ of $\mathbf{e}$ with respect to the d parameters $\mathbf{x}_t$, and what size does it have? **(1 point)**

   (b) A planar arm has two links of length 1 from a shoulder at the origin; $\mathbf{x} = (a_1, a_2)$ are the angle of link 1 and the angle of link 2 relative to link 1, so the link ends are $\tilde{\mathbf{r}}_1 = (\cos a_1, \sin a_1)$ and $\tilde{\mathbf{r}}_2 = \tilde{\mathbf{r}}_1 + (\cos(a_1 + a_2), \sin(a_1 + a_2))$; the observed ends are $\mathbf{r}_1 = (0.8, 0.6)$ and $\mathbf{r}_2 = (1.4, 1.4)$. From $\mathbf{x}_0 = (0, \pi/2)$ compute the residual $\mathbf{e}$ (predicted minus observed, in the order $e_{1x}, e_{1y}, e_{2x}, e_{2y}$), the Jacobian $\mathbf{J}$ (rows in that order, columns a_1, a_2), the Gauss-Newton step $\Delta\mathbf{x} = -(\mathbf{J}^T\mathbf{J})^{-1}\mathbf{J}^T\mathbf{e}$, and the error $\mathbf{e}^T\mathbf{e}$ before and after the step. **(2 points)**

4. **(3 points) Priors and regression.**

   (a) SMPLify minimises $E = E_J + \lambda_\theta E_\theta + \lambda_a E_a + \lambda_{sp} E_{sp} + \lambda_\beta E_\beta$ over the pose and the shape. (i) What does each of the four priors penalise? (ii) Under what assumption on the errors does a least-squares data term play the role of $-\log p(I \mid x)$, and what do the priors add so that minimising E gives a MAP estimate? **(2 points)**

   (b) HMR 2.0 regresses SMPL's parameters from the image with a vision transformer. What role does optimisation play at test time, and where does it come back in training? **(1 point)**

Question 6: Analysis by synthesis and neural mesh models (10 points)

1. **(3 points) Concepts.**

   (a) In one sentence each: how does a discriminator obtain the posterior $P(W \mid I)$, and how does analysis by synthesis obtain it? **(1 point)**

   (b) In render-and-compare, what does a neural mesh model compare with the image — rendered pixel colours, or something else — and why? **(1 point)**

   (c) In which three ways is a neural mesh model only an approximate analysis by synthesis? **(1 point)**

2. **(2 points) Figure 6: the two passes.** On the left of Figure 6 are the extracted features and the candidates they point to; on the right, three candidates and the features each predicts.

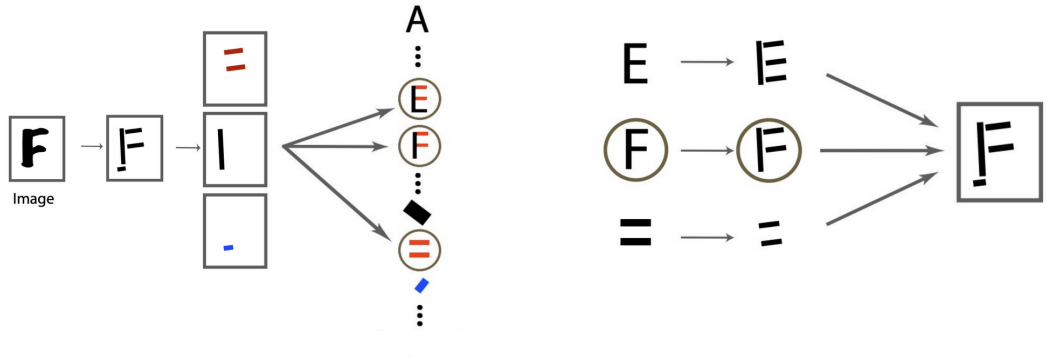


Figure 6: Analysis by synthesis on an image of an F.

(i) What does the bottom-up pass do, and why are only a few candidates kept for the second pass? (ii) What does the top-down pass do, and why is the F preferred to the E and to the parallel bars?

3. **(1 point) Inverse rendering.** Inverse rendering fits the parameters of a 3D object model by minimising a reconstruction loss between the rendering and the image. Which estimate, in the words of Bayes decision theory, does the loss alone give, and what must be added to it to find the most probable world state?

4. **(2 points) Render-and-compare on a small feature map.** A neural mesh model has three vertices with features $\theta_1 = (0, 1)$, $\theta_2 = (1, 0)$, $\theta_3 = (0, -1)$ and the clutter feature $\beta = (0, 0)$; all spreads are 1 and the constants are dropped, so the negative log-likelihood of a feature map under pose m is $\frac{1}{2}\sum_{i \in \mathcal{FG}} \|f_i - \theta_r\|^2 + \frac{1}{2}\sum_{i' \in \mathcal{BG}} \|f_{i'} - \beta\|^2$, r being the vertex rendered at position i. Five positions carry $f_1 = (0.2, 0.9)$, $f_2 = (0.8, 0.1)$, $f_3 = (0.1, -0.8)$, $f_4 = (-0.1, 0.2)$, $f_5 = (0.1, 0.1)$. Pose A renders vertices 1, 2, 3 at positions 1, 2, 3; pose B renders vertices 3, 2, 1 there; positions 4 and 5 are background under both. (i) Compute the negative log-likelihood of pose A and of pose B. (ii) Which pose does render-and-compare prefer, and by what factor is its likelihood larger?

5. **(2 points) Why analysis by synthesis is not widely used.** (i) The lecture says that analysis by synthesis needs object models that are hard to learn and hard to invert. What makes each of the two hard? (ii) GANs and diffusion models generate excellent images. Why are they not the generative models that analysis by synthesis needs?

Question 7: Detecting and parsing humans (7 points)

1. **(2 points) Concepts.**

   (a) Why does the method of this lecture combine a DCNN with a graphical model over the parts, instead of using the DCNN alone to detect the parts? **(1 point)**

   (b) What is an image dependent pairwise relation (IDPR), and what is a type of the relation between two neighbouring parts? **(1 point)**

2. **(2 points) Figure 7: the graph over the parts.**

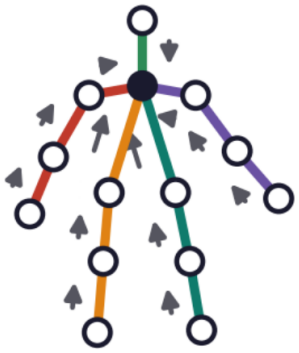


Figure 7: The model's graph drawn on a stick figure of 14 joints, with one node in black and arrows on its edges.

(i) Which node is the root, which nodes are the leaves, and in which direction do the messages $h_{j \to i}$ of dynamic programming travel? (ii) Why does dynamic programming find the pose with the largest score exactly on this graph?

3. **(2 points) Attach or cut.** In the flexible-composition model the message along the edge from the shoulder i to the elbow j either attaches the subtree below the elbow, with its message $h_{j\to i}(\mathbf{l}_i)$, or cuts it, with the score $B_{ij} + D_{ij}$, whichever is larger: $B_{ij} = \sum_k b_k$ adds a bias b_k for every removed part, and $D_{ij} = w_{ij}\,\varphi^d$ is the weighted DCNN log-probability that the elbow is occluded. Cutting removes the elbow and the wrist; take $w_{ij} = 1$, a DCNN probability of 0.5 that the elbow is occluded, and a best attached message $h_{j\to i}(\mathbf{l}_i) = -5.0$. (i) Compute the cut score when both removed parts have the bias $b_k = -1.5$, and again when both have $b_k = -3.0$; say in each case whether the arm is attached or cut. (ii) Why is the penalty B_{ij} needed — what would happen without it?

4. **(1 point) Why twice the time.** A chain of N parts has $N(N+1)/2$ flexible compositions, and a branching tree far more. Why can the best composition and pose nevertheless be found in only twice the time of the base model?

Question 8: Neural network generative models (7 points)

1. **(3 points) Concepts.**

   (a) A GAN is a game between two networks, the counterfeiter and the police. What is the input and what is the output of each network? **(1 point)**

   (b) StyleGAN's mapping network turns the Gaussian latent z into an intermediate latent $w = F(z)$, and it is w that sets the style of the generated image. Why is the space $\mathcal{W}$ of w more disentangled than the space $\mathcal{Z}$ of z? Use the lecture's example of a training set with no long-haired males. **(1 point)**

   (c) In mean field theory an intractable distribution P is approximated by a simpler, factorised distribution Q chosen to minimise a Kullback–Leibler divergence. In a variational autoencoder, which distribution plays the role of P and which the role of Q? **(1 point)**

2. **(2 points) Figure 8: Stable Diffusion.**

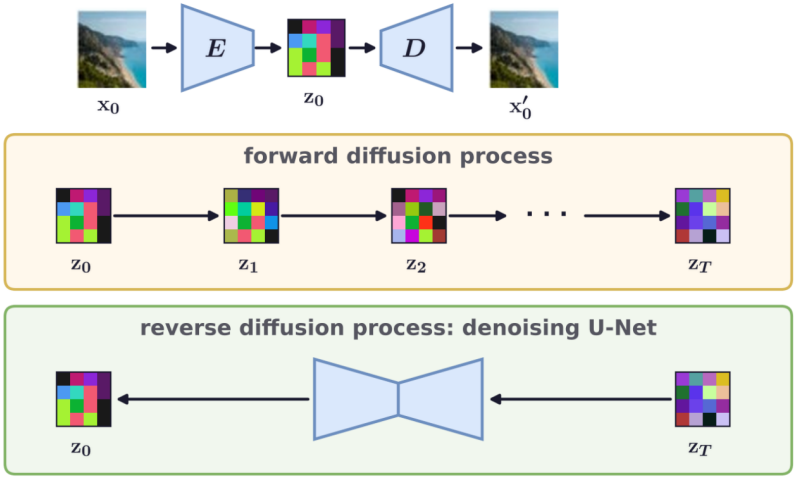


Figure 8: Stable Diffusion: an image x_0 through E and D (top), the forward (middle) and the reverse (bottom) diffusion process.

(i) E and D are the two parts of a network that is trained before the diffusion model (here D is not a discriminator). What network is it, and what are the coloured grids on which the two diffusion processes run? (ii) A 512×512 image with three colour channels enters E and leaves as a 64×64 array with four channels. By what factor does this cut the number of values each U-Net step processes, and why does Stable Diffusion perform diffusion in this latent space rather than in pixel space?

3. **(2 points) The optimal discriminator by hand.** For a fixed generator the discriminator maximises $V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{x\sim p_g}[\log(1-D(x))]$, and the best it can do is $D^*(x) = p_{\text{data}}(x)/(p_{\text{data}}(x)+p_g(x))$. Suppose there are only two kinds of image, A and B: the training data contain them with probabilities $p_{\text{data}} = (0.5, 0.5)$ and the generator produces them with probabilities $p_g = (0.8, 0.2)$. (i) Compute D^* for both kinds and the value $V(D^*, G)$. (ii) When $p_g = p_{\text{data}}$ the game has the value $-\log 4 = -1.386$, with $D^* = 1/2$ everywhere. Is your $V(D^*, G)$ above or below $-\log 4$, and why must it be so?

Questions 9–11: Experimental Section (38 points)

In this section you will run three Jupyter notebooks. They come with this PDF in the zip file of the homework; unzip it and keep the folder structure (the notebooks need no data files). The Python packages are listed in `environment.yml`; with conda, run `conda env create -f environment.yml` and `conda activate homework-3`, then start `jupyter notebook` in that folder. Each notebook runs in less than a minute on a laptop.

We only expect to see the generated plots and your answers to the questions in the notebooks, included in your PDF. Each question in a notebook is followed by a code cell for your experiments and a text cell for your answers; when a question asks for several parameter values, write a loop or copy the relevant cell into that code cell, so that every requested figure appears in your PDF.

Question 9: Geometry and motion: projection, epipolar lines and structure from motion (14 points)

Lecture 16. Notebook: `geometry_and_motion.ipynb`.
9.1 vanishing points, the horizon and scaled orthographic projection **(3)**; 9.2 epipolar lines from E and F **(4)**; 9.3 factorisation **(5)**; 9.4 non-rigid shapes **(2)**.

Question 10: Lambertian shading: photometric stereo, the bas-relief ambiguity and harmonic images (14 points)

Lecture 17 (the SVD: Lecture 16). Notebook: `lambertian_shading.ipynb`.
10.1 rank three, then not **(3)**; 10.2 photometric stereo, depth from normals and the bas-relief ambiguity **(5)**; 10.3 the Lambertian kernel and the nine harmonic images **(3)**; 10.4 fast marching: the distance transform and shape from shading **(3)**.

Question 11: Non-linear least squares, render-and-compare, and diffusion into noise (10 points)

Lectures 18 and 19 (the diffusion section: Lecture 15).
Notebook: `fitting_and_render_and_compare.ipynb`.
11.1 three optimisers on the toy arm **(4)**; 11.2 toy render-and-compare with an occluder **(4)**; 11.3 diffusing an image into noise, and back **(2)**.